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Intelligent Monitoring of Escape Routes via Computer Vision

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Abstract

[Insert thesis abstract in English here. This should provide a concise summary of the research problem, methodology, key results, and conclusions - typically 150-300 words.]

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Chapter 1

Introduction

1.1 Motivation

[Contextualize the problem and justify the work]

1.2 Research Question

[Clear statement of the research question]

1.3 Thesis Outline

[Overview of remaining chapters]

Chapter 2

State of the Art

2.1 Literature Overview

[Frame the work with respect to existing literature]

2.2 Research Question Reinforcement

[Highlight and reinforce the research question]

Chapter 3

Method

3.1 Objective

The goal of this work is to detect, on an edge device, when an obstacle blocks an evacuation route (a door or a corridor). Among the pipelines compared in this thesis, this chapter develops the third one, *Unsupervised Anomaly Detection with PatchCore* (P3), whose distinctive property is that it learns only from normal, unobstructed scenes and requires no labelled examples of obstructions. The following sections derive the components of the method, their mathematical formulation, and the data on which they operate; implementation choices and numerical settings are deferred to Chapter 4, and the empirical evaluation to Chapter 5.

3.2 Pipeline Overview

3.2.1 Unsupervised Anomaly Detection with PatchCore (P3)

Motivation and design rationale

The detection of emergency-exit occlusions presents a fundamental labelling bottleneck. Normal scenes - clear, unobstructed doorways and corridors - are easy to capture at scale, but the space of possible obstructions is open-ended: wheelchairs, carts, stretchers, waste bins, cardboard boxes, and any combination thereof can occupy a doorway in arbitrary configurations, orientations, and lighting conditions. Collecting and labelling a representative sample of anomalous scenes is time-consuming, and any finite labelled set risks leaving coverage gaps on obstacle types not seen during training.

P3 sidesteps this bottleneck by framing occlusion detection as *one-class anomaly detection*: the system learns only from normal scenes and, at inference time, flags any image whose feature representation deviates significantly from what was memorised as normal. This removes the need for labelled anomalous examples entirely. The choice to adapt PatchCore [1], originally proposed for industrial surface inspection,

to the emergency-exit domain is motivated by three properties that align with the deployment context:

1. **One-shot operation:** a single reference photograph of each monitored location is sufficient to build the entire anomaly model. In a real deployment, each fixed camera captures one reference frame of the clear exit; no further data collection is required.
2. **Spatial resolution:** PatchCore operates at the patch level rather than producing a single image-level embedding, preserving the spatial structure of the anomaly and enabling heatmap visualisation and spatially-aware filtering.
3. **Edge compatibility:** inference - feature extraction through a frozen backbone followed by a nearest-neighbour lookup in a compact memory - runs entirely on CPU, requires no gradient computation, and can be parallelised per camera without shared state.

Among the four pipelines, P3 is the most data-efficient: it needs only a single normal reference image and a pre-trained backbone, with no annotation effort to deploy on a new camera. The trade-off is that it cannot identify *which* obstacle is present - it only signals that something is anomalous - and it is, in principle, sensitive to any scene change, including non-obstacle changes such as lighting variations, shadows, or background movement. Managing these false-positive sources is the central methodological challenge of this chapter.

Key intuition

The core intuition is that the intermediate representations of a deep convolutional network pre-trained on a large natural-image corpus are broadly transferable: they capture generic visual patterns (textures, edges, object parts) that remain discriminative across domains, including surveillance footage of building interiors. A patch belonging to a normal scene of a fire-exit door yields a feature vector close to other normal patches in this embedding space, whereas a patch belonging to an unexpected obstacle - the wheel of a cart, the fabric of a stretcher, the surface of a box - yields a feature vector distant from all stored normal representations, hence a high anomaly score.

The memory is built from augmented variants of a single reference image rather than from a large dataset, exploiting the fixed-camera assumption: the camera does not move, so the background is stable across days and the only source of variation is illumination. Photometric augmentation at build time teaches the memory to tolerate this variation, making the decision threshold robust to normal fluctuations without requiring multiple observation days.

It is worth stressing why the alternative - populating the memory with a collection of *similar* exit scenes (many different corridors, many different doors) - would be both infeasible and counterproductive. Such scenes are visually too heterogeneous:

a memory mixing dozens of unrelated corridors and doorways would model the generic appearance of “a corridor” or “a door”, which is precisely the wrong target. Its feature manifold would be so broad that an obstacle in any specific scene could fall close to some unrelated normal scene in the bank and be scored as normal, destroying the detector’s sensitivity. The task is not to recognise corridors and doors in general, but to recognise *this* particular exit as it normally looks. This is the natural consequence of the deployment model, in which each camera monitors exactly one exit route: the normal class for that camera is its own fixed scene, and the only relevant variability is the illumination and the background change of that scene over time. A per-camera memory built from one reference image is therefore not a limitation imposed by scarce data, but the correct modelling choice for the objective.

Feature extraction

Let I_{ref} denote the normal reference image of a monitored location. Every image is resized to a fixed resolution and passed through a frozen convolutional backbone f . Following the original PatchCore formulation, features are read from two intermediate blocks, ℓ_2 and ℓ_3 : the shallower block preserves local detail while the deeper block carries semantic context, and their combination balances the two. The shallowest block encodes mostly low-level texture with little semantic content, and the deepest block is too abstract and biased toward the pre-training classes; both are therefore excluded.

The deeper feature map is upsampled bilinearly to the spatial resolution of the shallower one and concatenated along the channel axis, yielding a joint tensor of patch vectors arranged on a $h \times w$ grid. A local average pooling over a 3×3 neighbourhood (the neighbourhood aggregation of [1]) is then applied: each patch vector is replaced by the mean of its 3×3 spatial neighbourhood,

$$\phi(x) = \frac{1}{|\mathcal{N}_3(x)|} \sum_{x' \in \mathcal{N}_3(x)} \tilde{\phi}(x'), \quad (3.1)$$

where $\tilde{\phi}(x')$ is the raw concatenated feature at grid location x' and $\mathcal{N}_3(x)$ is the 3×3 window centred at x . This enlarges the receptive field of each vector and reduces sensitivity to pixel-level noise. The result is a set of $N_p = h \cdot w$ patch feature vectors per image.

Build phase

The build phase constructs the *memory bank* $\mathcal{M}$, the finite set of normal patch features against which every test patch will later be compared. It proceeds in three steps - augmentation, feature accumulation, and coreset reduction - each of which is detailed below. The guiding principle is that $\mathcal{M}$ must represent the full appearance of the normal scene under its expected illumination variation, yet remain small enough for a real-time nearest-neighbour search on an edge device.

The build phase is a *one-off, offline* operation, performed once when the camera is installed and its reference frame I_{ref} is captured. The resulting memory bank is then stored and reused unchanged for every subsequent inference; nothing in the build phase runs at monitoring time. A rebuild is required only when the *normal* reference scene itself changes in a way that is not a transient occlusion - for example a permanent rearrangement of the furniture in view, a repainting, or a change of the camera's mounting - so that the old I_{ref} no longer describes the normal state. Routine illumination changes and background variation over time are deliberately *not* a reason to rebuild: they are exactly what the augmentation and the calibration are designed to absorb. This separation between a rare, offline build and a continuous, lightweight inference is what makes the method practical on an edge device.

Step 1 - Augmentation. A single reference frame is too narrow a description of the normal scene: it captures one instant of illumination, whereas the same fixed camera will, over days, see the scene under brighter or dimmer light, warmer or cooler colour temperature, and varying sensor noise. To let the memory tolerate this variation, n_{aug} augmented variants are generated from I_{ref} , where n_{aug} is the number of bank variants. The augmentations are predominantly photometric - brightness, contrast, saturation, colour-temperature shifts, additive noise, and a mild blur - each motivated by a physical source of normal variation (time of day, light type, sensor noise, slight defocus). *Large* geometric transforms (flips, wide rotations, crops) are deliberately avoided: the camera is physically fixed, so orientation is semantically meaningful and must not be treated as a source of invariance, otherwise the memory would also accept an obstacle appearing in a flipped or arbitrarily relocated position as normal.

A controlled exception is a *small* geometric perturbation, off by default and enabled only for cameras subject to micro-vibration. When active, each variant is additionally shifted by a random translation of at most δ_{max} pixels on each axis and rotated by a random angle of at most ρ_{max} degrees, with δ_{max} of a few pixels and ρ_{max} of a couple of degrees. Its purpose is opposite to that of the forbidden large transforms: rather than making the memory invariant to where an obstacle sits, it makes the memory tolerant to a slight global displacement of the *whole* fixed scene, as produced by mechanical vibration or a small mounting drift. Such a displacement moves every patch by the same tiny amount, so a few-pixel jitter keeps a normal scene normal without ever spanning the distance an obstacle would cover. This is the cheapest of the available robustness measures, since it acts entirely at build time and adds no inference-time cost.

Step 2 - Feature accumulation. Each of the n_{aug} variants is passed through the feature pipeline of equation (3.1), producing N_p patch vectors per variant. Stacking the patches of all variants gives the *raw memory*

$$\mathcal{M}_{\text{raw}} = \bigcup_{v=1}^{n_{\text{aug}}} \{ \phi^{(v)}(x) : x \in \text{grid} \}, \quad |\mathcal{M}_{\text{raw}}| = n_{\text{aug}} \cdot N_p, \quad (3.2)$$

where $\phi^{(v)}(x)$ is the aggregated feature at grid location x of variant v . This mirrors the memory construction of [1], $\mathcal{M} = \bigcup_{x_i \in \mathcal{X}_N} \mathcal{P}(\phi(x_i))$, with the nominal training set $\mathcal{X}_N$ here instantiated by the augmented variants of the single reference I_{ref} . Each spatial location of the scene thus contributes n_{aug} nearby vectors - one per illumination variant - which together trace out the small region of feature space that a normal patch at that location can occupy. This is exactly the tolerance the detector needs: at test time a normal patch, under any illumination within the augmented range, will fall close to at least one of these vectors and receive a low anomaly score.

Step 3 - Coreset reduction. The raw memory is highly redundant: adjacent grid locations on flat surfaces (a wall, a floor, a door panel) and the n_{aug} illumination copies of each location produce many near-duplicate vectors. Storing all of them would inflate both the memory footprint and the nearest-neighbour cost without improving coverage. The memory is therefore compressed to a coreset $\mathcal{M} \subset \mathcal{M}_{\text{raw}}$ of size a fraction p of the original - the *coreset ratio* - that minimises the worst case distance of any raw vector to its closest retained vector [2, 1]:

$$\mathcal{M}^* = \arg \min_{\mathcal{M} \subset \mathcal{M}_{\text{raw}}} \max_{m \in \mathcal{M}_{\text{raw}}} \min_{m' \in \mathcal{M}} \|m - m'\|_2. \quad (3.3)$$

This minimax (facility-location) objective is NP-hard, so it is approximated by a greedy farthest-point procedure: starting from one random point, the algorithm repeatedly adds the still unselected vector that is *farthest* from the current selection,

$$m_t = \arg \max_{m \in \mathcal{M}_{\text{raw}} \setminus \mathcal{M}_t} \min_{m' \in \mathcal{M}_t} \|m - m'\|_2, \quad \mathcal{M}_{t+1} = \mathcal{M}_t \cup \{m_t\}. \quad (3.4)$$

Choosing, at each step, the point that maximises its minimum distance to the already-selected set spreads the retained vectors as evenly as possible over the occupied region of feature space: dense clusters of near-duplicates are summarised by a few representatives, while isolated points - which carry unique information about the normal scene - are kept. The procedure stops when $|\mathcal{M}| = \lceil p \cdot |\mathcal{M}_{\text{raw}}| \rceil$, subject to a small floor on the number of retained points so that even a very low p leaves a usable bank. Because evaluating equation (3.4) directly in the full feature dimension is expensive, the distance comparisons are performed on a low-dimensional random projection ψ that approximately preserves pairwise Euclidean distances (a Johnson-Lindenstrauss embedding [3]), exactly as in [1], while the selected vectors are stored at their original dimension. The result is a compact bank that fits in a few megabytes and enables sub-millisecond nearest-neighbour lookups, a prerequisite for edge deployment, while retaining near-uniform coverage of the normal-feature manifold.

Threshold calibration

Design space. Several threshold strategies are possible for a one-class detector: a fixed global value, a per-camera percentile, a ROC-optimal cut calibrated on labelled anomalies, or a parametric rule derived from the distribution of normal scores. A fixed global threshold requires that the absolute scale of anomaly scores be consistent across cameras, which does not hold in practice - a plain opaque door scores structurally lower than a glass door with a visible outdoor background, so a threshold tuned on one would misfire on the other. A ROC-optimal threshold requires labelled anomalies, violating the one-class premise and risking overfitting to the appearance of the synthetic anomalies used for tuning. A per-camera percentile is more sensitive to outliers in the small calibration sample. The chosen strategy is a parametric rule that adapts to each scene using only normal data.

Parametric rule. For each of n_{cal} additional augmented variants of I_{ref} (the *calibration variants*, disjoint from the bank variants and generated with an independent random seed), the maximum patch anomaly score over the image is computed. These n_{cal} scores form the empirical distribution of the maximum anomaly score of a normal scene, with mean μ and standard deviation σ . The decision threshold is

$$\theta = \mu + k \sigma, \quad (3.5)$$

where k is a sensitivity hyperparameter. This is the rule of statistical process control (Shewhart charts): an event is flagged when it lies k standard deviations above the normal mean. Under a Gaussian assumption, $k = 3$ corresponds to a theoretical false-positive rate of about 0.13% and $k = 4$ to about 0.003%; the calibration scores are not exactly Gaussian (they are maxima of correlated patch scores), but k still provides a monotone, interpretable control over the false-positive/recall trade-off. It is the sole deployment-time hyperparameter and encodes the asymmetric cost of a missed obstruction relative to a false alarm: for safety-critical exits k is set low (2.5 to 3.0) to favour recall, and higher (4.0 to 5.0) where false alarms are operationally costly.

The original I_{ref} is excluded from calibration as well: being the closest possible image to the memory (which is built from its augmentations), its score is structurally the minimum of the distribution rather than a representative sample, and including it would bias μ downward. Reserving I_{ref} as the actual normal test point - unseen by both the memory and the calibration - yields an honest estimate of the false-positive rate.

Coverage dependency. Because μ and σ are estimated on augmented variants, they reflect only the score variability those augmentations induce. If a real source of variation (cast shadows, light beams, outdoor background changes through glass) is absent from the augmentation scheme, σ is underestimated and real test scores under that variation exceed θ more often than expected. This dependency is the root

cause of the shadow false-positive problem (Section 5.1) and motivates the shadow and light augmentation introduced below.

Test phase and anomaly map

At inference, the query image is processed by the same feature pipeline to produce its patch vectors. Each patch vector ϕ is compared to the memory bank through nearest-neighbour search; let m^* be its nearest bank neighbour, so that the raw patch distance is $s^*(\phi) = \|\phi - m^*\|_2$. Let $\mathcal{N}_b(m^*)$ be the set of b nearest neighbours of m^* within the bank. Following equation 7 of [1], the patch score is re-weighted as

$$s(\phi) = w(\phi) s^*(\phi), \quad w(\phi) = 1 - \frac{\exp\|\phi - m^*\|}{\sum_{m \in \mathcal{N}_b(m^*)} \exp\|\phi - m\|}. \quad (3.6)$$

Two points deserve care. First, the distances in the denominator are between the test patch ϕ and the members of $\mathcal{N}_b(m^*)$, *not* between m^* and its own neighbours - a detail required for fidelity to the original formulation. Second, because m^* is the nearest neighbour, every distance in the denominator is at least $\|\phi - m^*\|$, so the fraction is at most $1/b$ and the weight is bounded as $w(\phi) \in [1 - 1/b, 1)$. The re-weighting therefore never inflates a patch score above its raw distance s^* : it leaves $s(\phi)$ close to s^* when m^* is isolated (its neighbours are far, the denominator is large, $w \rightarrow 1$), and discounts it by up to a factor $1 - 1/b$ when m^* sits in a dense region of common normal patterns (its neighbours are close, $w \rightarrow 1 - 1/b$). In relative terms this promotes patches matched to rare normal prototypes - the configuration most likely to indicate a genuine anomaly.

The original paper assigns the *image*-level score to the single most anomalous patch, $\arg \max_{\phi} s^*(\phi)$, re-weighting only that one (equations 6-7 of [1]). The present work instead evaluates equation (3.6) for *every* patch, arranges the scores on the spatial grid to form an anomaly map, upsamples it bilinearly, and smooths it with a Gaussian filter; the scalar image score is then the maximum of this smoothed map, $s_{\max}$, and the binary decision is $s_{\max} > \theta$. This is a deliberate variant of the paper’s single-patch score, adopted for two reasons: the full per-patch map is required by the spatial gating of Section 3.2.1, which operates on connected components of the map; and taking the maximum of the smoothed map - rather than of a single raw patch - is more robust to an isolated noisy patch, since smoothing broadens genuine anomaly blobs (helping small objects that span few patches) while attenuating one-pixel spikes. The two formulations coincide when no smoothing is applied and the maximiser of $w(\phi) s^*(\phi)$ is also the maximiser of s^* .

Shadow and light augmentation

The photometric-only memory contains no variants with cast shadows or localised light excess. This leaves a coverage gap: at test time, a shadow projected on the floor by a passer-by outside the field of view, or a light beam from a skylight or

a glass door, produces patch features distant from all bank entries and trips the threshold even though no obstruction is present. The two disturbances are symmetric - darkening versus brightening - and are addressed symmetrically by adding synthetic variants of each kind to the augmentation scheme. Three modalities are modelled for each (elliptical, directional, stripe), covering the physically plausible patterns in corridor and doorway scenes; all borders are blurred to avoid unrealistic high-frequency edges.

A disturbance is added to a build variant with probability p_{sh} (shadow) or p_{li} (light), and to a calibration variant with probability $p_{\text{sh}}^{\text{cal}}$ and $p_{\text{li}}^{\text{cal}}$ respectively. A central design choice is to *decouple* the build and calibration probabilities, because the two phases have opposing requirements:

- the **memory** needs maximum disturbance coverage so that test-time shadow/-light patches find near neighbours and produce low distances - hence a high build probability ($p_{\text{sh}} = p_{\text{li}} = 0.4$);
- the **calibration** needs a stable score distribution to yield a reliable μ and σ - hence a low calibration probability ($p_{\text{sh}}^{\text{cal}} = p_{\text{li}}^{\text{cal}} = 0.1$).

Setting the calibration probability equal to the build probability makes σ high-variance: a few unlucky calibration variants with heavy disturbances inflate σ , and the $k\sigma$ amplification produces per-camera thresholds that are either too high (false negatives) or too low (false positives). Decoupling removes this instability. The two families are independent: a variant may receive a shadow, a light, both, or neither. The disturbances are deliberately kept out of the test images used to measure robustness, so that the reported false-positive rates reflect generalisation to disturbances never seen at build time.

Spatial gating via annotated polygons

Glass doors introduce a distinct failure mode: the transparent panel makes the external background visible, and any movement or lighting change beyond the door generates anomaly blobs in the glass region without any physical obstruction. The principled remedy exploits a physical-grounding constraint: *any real obstruction rests on the floor and therefore intersects the door threshold*.

Door-frame polygons (left jamb, right jamb, central mullion, threshold, architrave) are annotated once on the reference image and rasterised into a binary *frame mask* F . The gating rule operates on the connected components of the thresholded anomaly map: the map is binarised at θ , its 8-connected components are labelled, and a component C is accepted only if

$$|C \cap F| \geq \tau, \quad (3.7)$$

where τ is the minimum overlap, in pixels, between a component and the frame mask. The final score is the maximum of the continuous anomaly map over the

accepted components, or zero if none survives. A component confined to the glass panel has zero overlap with F and is rejected, whereas a floor-grounded obstruction produces a component reaching the threshold or jambs and is accepted. The same rule applies to opaque doors when the panel exhibits strong illumination variation: specular highlights stay on the panel, while grounded obstructions reach the frame.

Structural limitations. Gating assumes the frame geometry is stable and visible at test time. An *open door* violates this: the threshold and jambs change radically, producing an out-of-distribution component inside F (a structural false positive), while an obstacle in the open doorway no longer intersects the misplaced mask (a structural false negative). A thin-stemmed grounded object (a floor fan, a sign on a pole) has only a narrow contact with the frame that may fail the overlap criterion τ . These cases share one root: reachability of the frame mask by a genuine anomaly component depends on frame visibility, a property of the scene rather than of the model. They are declared as limitations of the method; the system is designed and validated for the nominal closed-door condition, dominant in continuous surveillance.

Calibration stabilisation: pooled sigma

The per-camera rule $\theta_i = \mu_i + k \sigma_i$ has a structural instability. The per-camera standard deviation σ_i is estimated from a small calibration sample and carries a relative error on the order of 18%; this error propagates one-to-one into the $k\sigma_i$ term, and across many cameras it produces a long-tailed distribution of thresholds - a few cameras receive thresholds high enough to miss all obstructions, others low enough to generate shadow false positives, even though the underlying scenes are not fundamentally different. The extreme σ_i values are stochastic calibration outliers, not scene outliers: the same camera re-calibrated with a different seed would receive a normal threshold.

The fix replaces the per-camera σ_i with a single population estimate while keeping μ_i local:

$$\theta_i = \mu_i + k \sigma_{\text{pop}}, \quad \sigma_{\text{pop}} = \text{median}(\sigma_1, \dots, \sigma_N), \quad (3.8)$$

where N is the number of cameras. The median is robust to the outlier σ_i that caused the tail. Keeping μ_i per-camera preserves scene-specific adaptation (the absolute score level of each scene), while pooling σ removes the noisy multiplicative term. Under the per-camera rule $\text{Var}(\theta_i) = \text{Var}(\mu_i) + k^2 \text{Var}(\sigma_i) + 2k \text{Cov}(\mu_i, \sigma_i)$, whereas under the pooled rule it collapses to $\text{Var}(\theta_i) = \text{Var}(\mu_i)$, i.e. to the intrinsic scene variability alone.

Pooled sigma is, by construction, a shrinkage (regularisation) estimator: it trades a small bias - forcing each camera toward the population median - for a large variance reduction. In the data-starved regime of this thesis (a one-shot memory from a single image with sparse synthetic calibration) the per-camera σ_i is noise-dominated and the true cross-camera heterogeneity is near zero, so full pooling is the correct corner.

The principled generalisation is *partial pooling*, $\sigma_i^* = (1 - w_i) \sigma_{\text{pop}} + w_i \sigma_i$, with the weight w_i rising as a camera accumulates real normal frames; full pooling ($w_i = 0$) is the cold-start corner and remains valuable for newly installed cameras.

3.3 Dataset Description

[Description of the dataset used to build, calibrate, and evaluate the pipeline - to be completed.]

Chapter 4

Experimental Setup

4.1 Pipeline Implementation

4.1.1 Unsupervised Anomaly Detection with PatchCore (P3)

This section reports the concrete technical settings of the method derived in Chapter 3. Each monitored exit is handled by an independent memory bank built from a single reference frame, reflecting the deployment reality of a fixed camera with a stable normal scene.

Backbone and feature extraction

The backbone is a WideResNet50-2 [4] pre-trained on ImageNet and used frozen (no fine-tuning). Every image is resized to 224×224 and normalised with ImageNet statistics. Features are read from blocks ℓ_2 and ℓ_3 :

Block	Spatial resolution	Channels
ℓ_2	28×28	512
ℓ_3	14×14	1024

Table 4.1: Feature maps read from the frozen backbone at 224×224 input.

The ℓ_3 map is upsampled bilinearly to 28×28 and concatenated with ℓ_2 along the channel axis, giving a tensor of shape $[1536, 28, 28]$. The 3×3 neighbourhood average pooling of equation (3.1) is applied with unit stride and same-padding, and the result is flattened to $N_p = 784$ patch vectors of dimension 1536 per image. The shallowest block (56×56 , low-level texture) and the deepest block (7×7 , class-biased abstraction) are not used.

Build and calibration settings

The bank is built from $n_{\text{aug}} = 15$ photometric variants of I_{ref} (random seed 42); the original reference is excluded. The photometric augmentation ranges are listed in

Table 4.2.

Transform	Range	Rationale
Brightness	$\times [0.85, 1.15]$	Morning / evening / artificial light
Contrast	$\times [0.90, 1.10]$	Direct vs. diffuse illumination
Saturation	$\times [0.90, 1.10]$	LED / fluorescent / incandescent
Red-channel shift	$[-8, +8]$	Colour temperature
Blue-channel shift	$[-20, +20]$	Colour temperature
Gaussian noise	$\sigma \in [1, 4]$	Sensor noise
Gaussian blur	$r \in [0.5, 1.5], p = 0.3$	Slight lens defocus

Table 4.2: Photometric augmentation applied at build time. No large geometric transform is used, as the camera is fixed.

In addition to the photometric transforms, an optional small geometric perturbation is available for cameras subject to micro-vibration, disabled by default ($\delta_{\max} = 0$, $\rho_{\max} = 0$). When enabled, every variant receives a per-variant random translation drawn uniformly in $[-\delta_{\max}, +\delta_{\max}]$ pixels on each axis and a random rotation drawn uniformly in $[-\rho_{\max}, +\rho_{\max}]$ degrees, applied by bilinear affine warping at the original resolution; typical settings are $\delta_{\max} = 8$ px and $\rho_{\max} = 2^\circ$. Unlike the forbidden flips and wide rotations, this jitter perturbs the whole scene rigidly and teaches the memory to tolerate a small global displacement at no inference-time cost. It is left off for all experiments reported in Chapter 5, where the data-collection protocol (fixed foot position, braced camera) keeps hand-held variation within the native tolerance of the neighbourhood pooling and the anomaly-map smoothing.

The raw memory of $n_{\text{aug}} \cdot N_p = 15 \times 784 = 11760$ vectors is reduced by greedy coresets selection to a fraction $p = 0.01$ (retaining at least 50 points), leaving about 117 vectors. For efficiency the greedy selection of equation (3.4) runs on a sparse random projection to 128 dimensions (a Johnson-Lindenstrauss embedding), while the final coreset is taken from the original 1536-dimensional features.

The threshold is calibrated on $n_{\text{cal}} = 10$ further variants generated with seed 1000, disjoint from the bank seed to guarantee statistical independence; later experiments raise this to $n_{\text{cal}} = 30$ to reduce the sampling error on σ . The re-weighting of equation (3.6) uses a neighbourhood of size $b = 9$, and the anomaly map is smoothed with a Gaussian filter of $\sigma = 4$. The memory bank is serialised to a single file holding the coreset features together with the calibration parameters μ and σ .

Anti-leakage separation

The bank, calibration, and test sets are strictly disjoint (Table 4.3): no data used to build the memory or calibrate the threshold enters the test phase, and I_{ref} is reserved as the only genuinely unseen normal sample.

Set	Seed	Contains I_{ref} ?	Role
Bank variants ($n_{\text{aug}} = 15$)	42	No	Builds the feature memory
Calibration variants ($n_{\text{cal}} = 10$)	1000	No	Estimates μ and σ
Normal test	–	Yes	Independent normal sample
Obstructed test	–	–	Anomalous query

Table 4.3: Disjoint partition of data across build, calibration, and test.

Shadow and light augmentation

Both disturbance families implement three modalities, summarised in Table 4.4. They are applied after the photometric transforms, stacking $k_s \in [1, 3]$ instances per affected variant. The light family adds two axes absent from shadows: a diagonal (solar) stripe parameterised by a continuous incidence angle, and a warm colour shift (red up, blue down) on the elliptical and diagonal-stripe lights, modelling the chromaticity of direct sunlight. As a consequence the light subspace is intrinsically richer, so the raw values of light and shadow false-positive rates are not directly comparable; only within-category comparisons (across configurations) are legitimate.

Modality	Shadow	Light
Elliptical	dark ellipse, blurred border	bright ellipse (beam, specular)
Directional	dark edge gradient	bright edge gradient (lateral window)
Stripe	dark band (2 orientations)	bright band (3 orient., solar diagonal)
Colour shift	none	warm tint on elliptical / diagonal

Table 4.4: Synthetic disturbance modalities. Build probabilities $p_{\text{sh}}, p_{\text{li}} = 0.4$; calibration probabilities $p_{\text{sh}}^{\text{cal}}, p_{\text{li}}^{\text{cal}} = 0.1$.

To formally separate the test-time disturbances from the build-time ones, the build augmentation draws its shadows and lights from a dedicated random generator seeded disjointly from the generator that produces the test disturbances, so the two sets never coincide.

ROI annotation and connected-component gating

The frame mask of equation (3.7) is produced by a three-stage tool chain. A lightweight interactive annotator draws one polygon per frame element (left jamb, right jamb, central mullion, threshold, architrave, or a catch-all label) and writes one JSON file per image, storing polygons in original-image coordinates to avoid precision loss after resizing. An importer loads these polygons into the database, matching by relative file path. At evaluation time the polygon set of the reference frame is rasterised to the original resolution, the smoothed anomaly map is binarised at θ_i , its 8-connected components are extracted, and a component is kept only if its overlap with the frame mask reaches the minimum $\tau = 10$ pixels. The score is

the maximum of the continuous map over the surviving components, or zero if none survive.

The binarisation threshold used to form the components is the same per-camera θ_i used for the decision, so the sensitivity k acts at two stages - which pixels survive binarisation, and which final scores clear the decision. A legacy rectangular ROI can be composed in front of the polygon gating: the rectangle zeroes the map outside the door bounding box first (suppressing floor reflections and solar shadows), and the polygon rule then filters the remaining glass-panel variation. The full image is always included in the memory bank: excluding the glass region would leave its patches without any neighbour at test time, inflating their distance by mere absence of reference and making the panel anomalous by construction.

Pooled-sigma evaluation

The pooled threshold of equation (3.8) is computed in two passes over the cameras. The first pass builds each bank, calibrates μ_i and σ_i , and scores all test samples without yet deciding their label. The second pass collects the σ_i , sets σ_{pop} to their median, and recomputes for every sample the threshold $\theta_i = \mu_i + k \sigma_{\text{pop}}$, the normalised score $(s - \mu_i)/\sigma_{\text{pop}}$, and the binary decision. The normalisation makes scores comparable across cameras of different absolute scale, so that the aggregated AUROC measures intrinsic separability rather than scale differences. The mode (per-camera or pooled) and the value of σ_{pop} are recorded with the experiment for reproducibility. Pooling affects neither inference time nor memory size, since calibration variants never enter the coreset; its only cost is a one-off computation at build time.

4.2 Application

[General description of the application, its modules, and overall architecture - to be completed.]

Chapter 5

Experimental Analysis

5.1 Pipeline Results

5.1.1 Unsupervised Anomaly Detection with PatchCore (P3)

Evaluation protocol

All results are obtained through a database-driven evaluation that, for each normal reference frame, (i) builds the memory bank from augmented variants of the reference (excluding the original), (ii) calibrates the threshold on independent variants, (iii) scores the original reference as the normal test sample, and (iv) scores the obstructed frames linked to that reference. Scores are normalised per camera as $(s - \mu_i)/\sigma_i$ before aggregating the AUROC, while threshold-based metrics (false-positive rate FPR, true-positive rate TPR, and F_1) use the per-camera threshold θ_i so that each binary decision reflects its own calibration. Unless stated otherwise, experiments run on the copy-paste synthetic dataset (split *test*, doors and corridors together), about 990 reference frames each with one or more synthetic obstructed variants.

The evaluation is organised as a progressive ablation, in which each component is introduced in response to a failure mode identified empirically (Table 5.1).

Phase	Configuration	Purpose
A	Baseline (no disturbances, no ROI)	Characterise base behaviour
B	Baseline + shadow test set	Quantify shadow FPR
C	Shadow augmentation + pooled σ	Fix shadow FPR, stabilise θ
D	Real doors + polygon gating	Validate full pipeline, recalibrate k
E	Light augmentation ablation	Reduce light FPR; separate coverage from budget

Table 5.1: Progressive-ablation structure of the evaluation.

Phase A: baseline on the copy-paste dataset

The baseline ($n_{\text{aug}} = 15$, $n_{\text{cal}} = 10$, no disturbance augmentation, $k = 3$) cleanly separates normal from obstructed scenes (Table 5.2). The near-perfect AUROC and the gap of several standard deviations between the class means confirm that obstacle patches fall well outside the distribution of normal door and corridor patches. The residual 0.5% FPR corresponds to about five false alarms over 990 frames. This establishes the detector as viable and validates the evaluation methodology.

Category	Mean score	Std	Rate
Normal	2.309	0.156	FPR 0.5%
Obstructed	4.648	0.348	TPR 100%
AUROC	≈ 1.000		

Table 5.2: Phase A baseline on the synthetic test set.

Phase B: shadow vulnerability

A test set of shadow-augmented normal scenes (one synthetic variant per reference, $k_s \in [1, 3]$ stacked shadows) is scored against the baseline model. These are normal scenes with no obstruction, yet 90.1% are flagged as anomalous (Table 5.3). Shadow patches - dark, smoothly-bordered regions - produce features unlike any bank entry, which contains only shadow-free photometric variants; the baseline cannot distinguish a cast shadow from a physical obstruction at the feature level.

Category	Mean	Std	Range	Rate
Normal	2.309	0.156	[1.87, 3.13]	FPR 0.5%
Shadow normal	3.189	0.495	[2.10, 4.51]	FPR 90.1%
Obstructed	4.648	0.348	[3.54, 5.87]	TPR 100%

Table 5.3: Phase B: shadow-normal scores overlap the obstructed range [3.54, 4.51].

The distributions overlap in [3.54, 4.51], so no fixed global threshold achieves acceptable shadow FPR and TPR simultaneously: at a cut of 4.00 the shadow FPR is 4.8% but TPR has already dropped to 97.3%, and pushing the cut to 4.50 to reach 0.1% shadow FPR collapses TPR to 67.4%. The problem is not threshold calibration but a distribution mismatch between the shadow-free memory and the shadow-bearing test set.

Phase C: shadow augmentation and pooled sigma

Introducing shadow augmentation in the build phase, then decoupling and pooling the calibration, traces a clear path (Table 5.4). Coupling the build and calibration probabilities ($p_{\text{sh}} = p_{\text{sh}}^{\text{cal}} = 0.4$) reduces shadow FPR by two orders of magnitude but over-inflates the threshold, dropping TPR to 70.6%; the cause is a long tail in

the per-camera threshold distribution, where 12% of cameras generate 86% of all false negatives. Decoupling ($p_{\text{sh}}^{\text{cal}} = 0.1$, $n_{\text{cal}} = 30$) recovers TPR but swings to an over-permissive operating point. Replacing the per-camera σ_i with the pooled median $\sigma_{\text{pop}} = 0.300$ at $k = 4$ eliminates the residual threshold variance and reaches the best operating point.

Configuration	Clean FPR	Shadow FPR	TPR	F_1	σ_θ
Baseline (no aug)	0.5%	90.1%	100%	0.688	0.158
Coupled ($k = 3$)	0.0%	2.1%	70.6%	0.818	0.539
Decoupled ($k = 3$)	0.0%	28.4%	99.4%	0.873	0.388
Decoupled ($k = 4$)	0.0%	13.5%	93.4%	0.903	0.497
Pooled σ ($k = 4$)	0.0%	12.2%	99.8%	0.941	0.137

Table 5.4: Phase C: shadow augmentation and calibration stabilisation. σ_θ is the standard deviation of the per-camera thresholds.

Pooling collapses the threshold range from [2.28, 6.09] to [3.24, 4.16] (a 73% reduction in σ_θ) and removes the pathological tail entirely: no camera exceeds $\theta = 4.16$ and TPR stays at 99.3% or higher in every threshold bucket. The 3.8 percentage-point F_1 gain is due entirely to variance reduction, not to any change in the backbone, whose AUROC stays at 0.998. The choice $k = 4$ follows a conservative four-sigma rule and was not tuned on the test set: the F_1 -optimal value is about 3.9, but reporting it would constitute test-set tuning.

Phase D: real-world validation on real doors

Phase D is the first evaluation on real photographs rather than copy-paste composites. The test set consists of six real doors photographed on site (a mix of opaque and glass doors), each with one annotated reference frame, comprising 30 real obstructed images, 30 unobstructed images, and the six references. Consistent with the fixed-camera assumption, all shots of a door share the same framing, so the negatives measure the FPR under normal content and illumination variability at fixed geometry. The full pipeline is active: pooled σ plus connected-component gating ($\tau = 10$).

Validated on the synthetic suites, $k = 4$ proved too conservative on real scenes; lowering it to $k = 3$ raises recall at no FPR cost (Table 5.5). Because the pooled median(σ) = 0.213 is a shared constant, lowering k shifts every threshold down by the same amount without widening the distribution - the per-camera variance amplification of the naive rule does not reappear.

Comparing the two runs separates the false negatives by cause. Two are *calibration-limited* and are recovered at $k = 3$ through the two distinct stages where k acts: one is a decision-stage miss (a valid gated component scored just below the decision threshold at $k = 4$), the other a binarisation-stage miss (the obstruction’s anomaly peak fell just below the local binarisation threshold at $k = 4$, leaving an empty bi-

Metric	$k = 4$	$k = 3$	Δ
Recall (TPR)	83.3%	90.0%	+6.7 pp
FPR	3.3%	3.3%	invariant
Precision	96.2%	96.4%	+0.2 pp
F_1	0.893	0.931	+0.038
Accuracy	90.0%	93.3%	+3.3 pp

Table 5.5: Phase D: recalibration of k on the real test set.

nary map). The latter shows that θ controls not only the decision but which pixels exist for gating to evaluate, so an overly high k can erase a low-contrast obstruction before gating sees it. Three remain zeroed even at $k = 3$ and are *gating-limited*: the prototypical case is a free-standing floor fan whose wide head sits on the door panel (excluded from the frame mask by design) and whose only frame contact, the thin pole, narrows to 1-2 px after resizing and fails the $\tau = 10$ overlap. PatchCore detects the fan correctly; the failure is purely geometric, confirming the structural limitation anticipated in Chapter 3: the grounded-obstacle assumption holds for wide-profile objects (carts, wheelchairs, bins) but not for thin-stemmed ones (fans, coat racks, sign posts). A single FP is a glass door whose background changed behind the panel. With order-of-tens samples, these numbers are a qualitative validation of the complete pipeline, not a production benchmark.

Phase E: light augmentation ablation

Phase E addresses the symmetric counterpart of the shadow failure mode - localised brightness excess - evaluated on a light-augmented test set (≈ 1100 normal scenes) at $k = 4$, pooled σ , $n_{\text{cal}} = 30$. Five configurations are compared (Table 5.6), where the *budget* $p_{\text{sh}} + p_{\text{li}}$ is the expected disturbance probability per variant, a proxy for total augmentation intensity. A fifth fair-budget control (L4) disentangles two effects that L2/L3 confound: increased coverage (adding the light type) and increased total intensity.

Config	p_{sh}	p_{li}	Budget	Shadow FPR	Light FPR	F_1
L0 shadow-only	0.4	0.0	0.40	12.2%	5.8%	0.917
L1 light-only	0.0	0.4	0.40	37.7%	23.2%	0.767
L2 SL-15	0.4	0.4	0.80	2.2%	0.30%	0.982
L3 SL-25	0.4	0.4	0.80	1.1%	0.10%	0.982
L4 fair-budget	0.23	0.23	0.46	11.2%	4.6%	0.929

Table 5.6: Phase E light-augmentation ablation. TPR is $\geq 97.7\%$ throughout. L0/L2 use $n_{\text{aug}} = 15$, L3 uses $n_{\text{aug}} = 25$.

Four findings emerge. First, *light is a milder disturbance than shadow*: even with no light coverage (L0) the light FPR is 5.8%, below the 12.2% shadow FPR, because

a light beam pushes features less out of distribution than a cast shadow. Second, *adding light augmentation works and helps shadows too*: from L0 to L2 the light FPR falls about nineteen-fold while the shadow FPR also improves rather than degrading. Third, L1 confirms that shadow augmentation is *structural, not incidental*: removing it collapses the per-camera threshold and inflates all FPRs, light included - light augmentation is a complement to shadow augmentation, not a substitute. Fourth, the fair-budget control shows that *the headline gain is mostly a budget effect*: L2/L3 apply twice the augmentation of L0, and the mean threshold grows monotonically with budget, tracking the FP reduction; at equal budget (L4 vs. L0) diversification alone yields only a small, cost-free improvement. The recommended configuration is SL-15 ($F_1 = 0.982$, light FPR 0.30%, shadow FPR 2.2%, recall 99.0%); SL-25 halves the FPRs again but trades recall for identical F_1 at a markedly longer build time, justified only when minimal FP is required. As established in Chapter 4, light and shadow FPR are not directly comparable; only within-category readings are valid.

Discrimination versus calibration

A cross-cutting question determines what the pipeline can legitimately claim: do the interventions improve the model’s *discriminative capacity* or only the *placement of the threshold*? The clean metric is the per-camera AUROC - the probability that an obstruction scores higher than a normal variant of the *same* camera, on raw scores, averaged over cameras. Within a camera μ_i and σ_i are constant, so raw and normalised scores differ by an affine transform that preserves ranking; the per-camera AUROC is therefore invariant to both threshold and normalisation and isolates intrinsic discrimination.

Config	raw AUROC	per-cam AUROC (shadow)	shadow FPR	TPR	F_1
no-aug	0.9969	0.996	90.1%	100%	0.688
shadow04	0.9981	0.996	2.1%	70.6%	0.818
decoupled	0.9981	0.996	28.4%	99.4%	0.873
decoupled $k = 4$	0.9981	0.996	13.5%	93.4%	0.903
pooled	0.9981	0.996	12.2%	99.8%	0.941

Table 5.7: Controlled comparison: shadow FPR swings 40× while raw and per-camera AUROC stay fixed.

The per-camera AUROC is saturated and nearly invariant to augmentation (Table 5.7): clean separation is perfect everywhere, the shadow value stays at 0.996 across the whole path - even with no shadow augmentation - and light improves only from 0.997 to 1.000. The shadow FPR meanwhile swings from 2.1% to 90% to 12% while the raw AUROC is fixed at 0.9981. The mechanism is a *distributional shift, not a re-ordering*: without shadows in the memory, shadow patches have no neighbours and score high, staying below the obstructed scores (ranking preserved) but above a threshold calibrated on clean scenes (FPR 90%); with shadows in the

memory, their scores drop below the threshold without changing order relative to obstructions. Augmentation shifts the disturbed-normal distribution downward relative to a fixed threshold; pooled σ then stabilises where that threshold lands across cameras. Neither adds discriminative power - they convert an already-excellent latent discrimination into a usable single operating point.

This conclusion holds on a test set where discrimination is near-saturated by design (copy-paste obstructions on the same background: only the object is anomalous, a strong signal, AUROC ≈ 1). On harder data - real obstructions, variable backgrounds - discrimination would have headroom and augmentation might contribute to separability as well; the per-camera AUROC argument is a claim about this dataset's regime and is stated as such.

Discussion

The progressive ablation shows that the failure modes of the method are diagnosable and fixable within the PatchCore framework without architectural changes. Cast shadows are addressed by synthetic shadow variants in the memory with decoupled calibration (shadow FPR 90.1% $\rightarrow$ 2.2% at SL-15); localised brightness excess is addressed symmetrically by light variants; glass-door backgrounds are addressed by connected-component gating exploiting the floor-grounding constraint; and threshold instability across cameras is addressed by pooling the σ estimate. The unifying finding is that none of these fixes change the backbone's discriminative capacity, which is saturated; the bottleneck is placing a single threshold robust to disturbances.

The conclusions carry two tiers of statistical weight. The synthetic phases ($N \approx 1000$) are quantitative validations with adequate power: the measured deltas of shadow augmentation, decoupling, and pooled σ are stable and generalisable to similar synthetic scenes. The real-door phase ($N \approx \text{tens}$) is a qualitative validation of the complete pipeline: it confirms the expected behaviour of gating plus pooled σ and motivates the k recalibration, but the absolute numbers are illustrative, not a production benchmark. The structural results - augmentation effectiveness, decoupling rationale, pooled-sigma variance reduction, and the discrimination-versus-calibration mechanism - are pipeline properties that generalise; the exact operating-point metrics should be recalibrated for any new deployment using a site-specific σ_{pop} and a k chosen from the application's cost ratio of false negatives to false positives. Larger-scale quantitative validation of the gating on real glass doors, conditioned on a semi-automatically annotated test set, is the natural next step.

5.2 Application Results

[Analysis of the end-to-end application results - to be completed.]

Chapter 6

Conclusions

[Summary and future work]

Appendix A

Questa è un'appendice

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Bibliography

- [1] Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter Gehler. Towards total recall in industrial anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14318–14328, 2022.
- [2] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations (ICLR)*, 2018.
- [3] Wikipedia contributors. Johnson–lindenstrauss lemma — Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Johnson%E2%80%93Lindenstrauss_lemma&oldid=1357719950, 2026. [Online; accessed 10-June-2026].
- [4] Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2016.